

Project Report

Bank Marketing Campaign — Predicting Term Deposit Subscription
AI-Assisted Data Analytics Final Project — Malak Usama

1. Dataset Overview

This project uses the **UCI Bank Marketing Dataset** (bank-additional-full), a real-world record of 41,188 telemarketing contacts made by a Portuguese retail bank between 2008 and 2010, across 21 variables. The dataset sits in the **retail banking / direct marketing / finance** domain.

Business problem: outbound telemarketing for term deposits is expensive per call, yet only roughly 11% of contacted clients subscribe. The objective is to predict, before a client is called, how likely they are to subscribe — so the contact center can prioritize its limited calling capacity toward the highest-propensity clients, raising conversion per call and lowering acquisition cost.

Target variable: y — whether the client subscribed to a term deposit (yes/no). This is a clean, unambiguous binary label present on every row, with enough positive examples (~4,640 subscriptions, 11.3% of records) to support supervised classification without being impractically rare.

Project objectives: clean and prepare the raw data, explore it for business-relevant patterns, engineer features, train and compare multiple classification models, define and evaluate a KPI against a target, and communicate the results through an interactive Tableau dashboard.

2. Data Preparation

Data quality issues identified

- 12 exact duplicate rows.
- "unknown" placeholders standing in for missing values across six categorical columns — most notably *default* (20.9% missing), *education* (~4.2%), *housing/loan* (~2.4% each), *job* (~0.8%), and *marital* (~0.02%).
- A non-numeric sentinel value (999) inside *pdays*, meaning "never contacted before" rather than a real day count.
- Right-skewed outliers in *campaign* (up to 56 contacts) and *duration* (calls over an hour).

Cleaning process

- Removed all 12 duplicate rows.
- For low-missingness columns (*job*, *marital*, *education*, *housing*, *loan*), imputed "unknown" with the column mode.
- For *default*, kept "unknown" as its own explicit category rather than imputing, since 20.9% missing is too much to safely fill and a client's non-disclosure is itself informative.
- Converted the *pdays* sentinel into an explicit *was_contacted_before* flag plus a cleaned distance value, avoiding a fake "999 days" from distorting any model.
- Winsorized *campaign* and *duration* at the 99th percentile rather than deleting outlier rows, preserving genuine (if rare) client behavior.

Feature engineering

- *age_group* — bucketed age into five business-interpretable bands (18-30, 31-40, 41-50, 51-60, 60+).

- *duration_min* — call duration converted to minutes for interpretability.
- *prior_success* — binary flag for whether a previous campaign already succeeded with this client.
- *quarter* — contact month rolled up to fiscal quarter for coarser seasonality analysis.
- *total_contacts* — sum of this campaign's and all previous campaigns' contact counts.

AI-assisted preprocessing

Generative AI was used to help identify the cleaning strategy for each column (e.g., recommending mode-imputation only where missingness was low, and flagging that *default*'s 20.9% missingness was too high to impute safely) and to flag *duration* as a data-leakage risk before modeling. All recommendations were verified against the actual column statistics before being applied.

3. Exploratory Data Analysis

- **Class imbalance:** subscriptions are rare — 11.27% yes vs. 88.73% no — meaning accuracy alone is a misleading evaluation metric for this problem.
- **Age:** subscription rate is U-shaped — clients 60+ (45.5%) and, to a lesser extent, 18-30 convert far above the 31-50 core working-age segment (roughly 8-15%), likely reflecting different savings priorities (retirement income vs. building initial savings) versus mortgage-carrying mid-career clients.
- **Job:** students (31.4%) and retirees (25.3%) convert at 2-3x the rate of blue-collar (6.9%) or services (8.1%) workers, consistent with the age pattern.
- **Seasonality:** conversion swings widely by month — lower-volume months (March 50.6%, September, December 48.9%) convert far better than the high-volume summer campaign push, suggesting the bank's biggest campaign waves are timed at the least efficient point in the year.
- **Prior campaign outcome:** a past campaign success is the single strongest categorical signal in the data — clients who said yes before are dramatically more likely to say yes again.
- **Economic conditions:** higher Euribor 3-month rates and stronger employment both correlate with lower subscription rates — term deposits compete less well when the economy is running hot.
- **Outlier/leakage note:** median call duration for a "yes" is roughly triple that of a "no" — a strong signal, but only available after the call ends (see Section 4).

4. Machine Learning

Four algorithms were implemented and compared: **Logistic Regression**, **Decision Tree**, **Random Forest**, and **Linear SVM**. Numeric features were scaled with a **RobustScaler** (median/IQR-based) rather than standard scaling, since several features remain right-skewed even after outlier capping. Categorical features were one-hot encoded. All models used balanced class weighting to account for the ~89/11 class imbalance.

Leakage-aware modeling design

The *duration* column is only known after a call has already ended, so using it would be data leakage in a real deployment scenario — the bank cannot know how long a call will last before making it. To be rigorous, every model was trained twice: once **with duration** (a benchmark showing the theoretical performance ceiling) and once **without duration** (the honest, deployable version usable to decide who to call before dialing).

Model comparison results

Feature Set	Model	Accuracy	Precision	Recall	F1	ROC-AUC
With duration	Random Forest	0.885	0.495	0.878	0.633	0.946
With duration	Logistic Regression	0.865	0.450	0.905	0.601	0.940
With duration	Decision Tree	0.855	0.431	0.911	0.585	0.925
With duration	SVM (Linear)	0.910	0.646	0.438	0.522	0.940
Without duration	Random Forest	0.872	0.448	0.601	0.513	0.812
Without duration	Logistic Regression	0.829	0.356	0.643	0.459	0.800
Without duration	Decision Tree	0.826	0.351	0.641	0.454	0.780
Without duration	SVM (Linear)	0.897	0.629	0.203	0.306	0.800

Best-performing model and justification

Random Forest is the selected final model on the deployable (no-duration) feature set, achieving the best F1 (0.514) and ROC-AUC (0.812) among the four algorithms. It captures non-linear interactions between economic indicators, prior-contact history, and demographics that a single linear model (Logistic Regression) or a shallow single tree (Decision Tree) cannot. Linear SVM is competitive on ROC-AUC but weaker on F1/recall. The roughly 15-20 point ROC-AUC gap between the with-duration benchmark and the deployable model is expected, since duration is a symptom of client interest, not a cause — the deployable model has to predict that interest before the symptom exists, which is an inherently harder problem.

5. Dashboard Summary

The interactive dashboard was built in **Tableau** as three linked tabs: *Overview & KPI*, *Deep-Dive Drivers*, and *Model Performance*, connected with navigation buttons so the viewer can move between a summary view and progressively deeper detail.

- **Overview & KPI** — the headline KPI tile, target distribution, subscription rate by age group, and by job, with *job* and *age group* filters applied across all charts on the tab.
- **Deep-Dive Drivers** — seasonality (rate by month), economic conditions vs. subscription, and prior campaign outcome, with *poutcome* and *month* filters.
- **Model Performance** — a confusion-matrix-style prediction accuracy grid, feature importance ranking, and a full model comparison table, with a *model* filter.

Design choices: a consistent pale-green/dark-green color theme applied across every sheet, mark labels shown on every chart for direct readability, and a one-line plain-language insight caption placed under each chart so the dashboard reads as a narrative rather than a disconnected set of charts. Predictive insight is surfaced directly in the Model Performance tab (feature importances and the prediction-accuracy breakdown) rather than only in the notebook.

6. KPI Explanation

KPI: Top-20% Model Lift

Definition: the ratio of the subscription rate among the top 20% of clients ranked by the model's predicted probability, divided by the overall baseline subscription rate across all clients.

Target value: 2.0x lift.

Target justification: a 2x lift on the top-scored quintile is a commonly used minimum bar in response-modeling practice for "the model is worth operationalizing" — below that threshold, the model doesn't clearly outperform a simple rule-based heuristic (e.g., "call everyone with a prior campaign success"), so 2.0x was chosen as a concrete, defensible efficiency bar rather than an arbitrary number.

Performance vs. target: the model achieved a **3.52x lift** — the top-scored 20% of clients convert at 39.68%, versus an 11.27% baseline conversion rate across the full client population. **The target was met**, and by a wide margin: the bank could reach roughly the same number of conversions while contacting a substantially smaller share of its client list, or convert far more clients for the same number of calls.

7. AI-Generated Insights

(Generated with AI assistance, then reviewed and verified against the EDA and model results before inclusion below.)

Key trends

- Prior relationship with the bank is the dominant driver of conversion — a past campaign success and having been contacted before at all are far stronger signals than any demographic attribute.
- Macroeconomic context matters as much as who is called — campaigns launched when Euribor and employment variation are low convert meaningfully better, a lever the bank controls independently of targeting.
- Life-stage segments at the edges of the working-age range (students, retirees, 60+) convert at roughly double the rate of core working-age segments.

- The bank's current campaign calendar is weighted toward its lowest-converting months.

Interpretation of model performance

The ROC-AUC gap between the benchmark (with duration) and deployable (without duration) models is expected: duration is a proxy for "the client was engaged enough to keep talking," a symptom of interest rather than a cause of it. The deployable model's task — predicting interest before that symptom exists — is inherently harder and noisier, which is reflected in its lower but still meaningfully-better-than-baseline performance. Because the classes are imbalanced, F1 and recall on the minority class are the metrics that matter, not raw accuracy.

Business implications

- Prioritizing clients with a prior campaign success, prior contact history, or student/retired status would let the contact center concentrate effort on the highest-propensity clients.
- Shifting call volume toward historically higher-converting months could raise overall conversion without added budget.
- Preferring cellular contact where available should mechanically lift response rates.

Actionable recommendations

- Integrate the deployable Random Forest propensity score into the dialer/CRM so agents call high-probability clients first each shift.
- Re-balance campaign volume across the calendar toward historically stronger-converting months.
- Route call attempts to cellular numbers first, falling back to landline only when no mobile is on file.
- Create a lighter-touch "warm" queue for clients with a prior campaign success or recent prior contact, since they convert at a multiple of the base rate.

8. Final Conclusions

This project demonstrates the full analytics-to-decision workflow on a real banking dataset: cleaning genuine data-quality issues, uncovering business-relevant patterns through EDA, building and honestly comparing four classification models under a leakage-aware design, and translating the results into a KPI-driven, interactive dashboard. The dominant conversion drivers are relationship history and macroeconomic timing, not demographics alone. The deployable Random Forest model, used to prioritize the top 20% of clients by predicted probability, delivers a 3.52x lift over untargeted calling — well above the 2.0x target — giving the bank a concrete, actionable tool for raising campaign efficiency. Recommended next steps include hyperparameter tuning, testing resampling strategies (e.g., SMOTE) against the current balanced class-weighting approach, and periodic retraining as macroeconomic conditions shift quarter to quarter.